

Effect of dietary antioxidants on postprandial endothelial dysfunction induced by a high-fat meal in healthy subjects.

BACKGROUND: Persons following current dietary guidelines have a lower risk of mortality from coronary heart disease. **OBJECTIVE:** The objective was to compare the short-term effect of a high-fat meal and a high-carbohydrate meal, with and without dietary antioxidants, on vasomotor, antiplatelet, and hemostatic functions of the endothelium in healthy subjects. **DESIGN:** In an observer-blinded, randomized crossover study, 25 (13 men and 12 women) healthy subjects were given each of 3 meals in random order at 1-wk intervals: a high-fat meal (760 kcal), an isoenergetic high-carbohydrate meal, and a high-fat meal with dietary antioxidants from vegetables (865 kcal). Endothelial functions, as assessed by hemodynamic and rheologic responses to L-arginine--the natural precursor of nitric oxide--were evaluated before and 4 h after each meal. **RESULTS:** Unlike the high-carbohydrate meal, the high-fat meal increased the plasma concentrations of triacylglycerol ($P < 0.01$); both meals activated hemostasis. The high-carbohydrate meal did not modify blood pressure, and platelet aggregation decreased in response to the L-arginine load (-7.1 ± 2.3 mm Hg and $-8.5 \pm 4.5\%$, respectively). After the high-fat meal, the decrease in blood pressure in response to L-arginine was reduced (-1 ± 0.8 mm Hg), and platelet aggregation showed a paradoxical increase ($4.1 \pm 2.1\%$; $P < 0.01$ compared with the high-carbohydrate meal). The high-fat meal with antioxidants partially restored the vascular response to L-arginine. **CONCLUSION:** Compared with a high-carbohydrate meal, a high-fat meal can modify endothelial functions toward a more atherogenic profile, which is partially prevented by dietary antioxidants.